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Development of a robust data-driven surrogate model to improve energy flexibility of an integrated district heating system with a thermal storage system

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ABSTRACT

Data-driven surrogate models (SMs) are increasingly applied to optimize energy systems, yet the impact of dataset variability and diversity on model robustness is often overlooked in operational optimization. Phase change material (PCM) tanks can enhance energy flexibility of district heating (DH) systems, but an integrated strategy that considers both spot price fluctuations and peak demand remains underexplored. To bridge this research gap, this study presents a robust SM to enhance the performance of a DH system integrated with a data centre and a PCM thermal storage system. Various operational strategies for PCM tank charging and discharging were implemented to generate diverse datasets for training the SMs and identifying the most robust model. Long short-term memory networks were used to develop the models, following careful input-output selection. A new operational strategy was then proposed, focusing on peak shaving and load shifting, using the best-performing SM to optimize tank operation. A case study showed that the day-ahead price-based control strategy improved model robustness by enabling dynamic day-to-day adjustments in charging and discharging, leading to more effective management and the generation of well-distributed datasets. The tailored strategy achieved an 11.5 % reduction in peak demand, a 0.3 % decrease in price-responsive heating costs, and a 6.1 % reduction in total operational costs compared with the daily time-based strategy. This study provides a reference for developing robust SMs using datasets generated by demand response strategies and offers guidance on improving the performance of hybrid DH systems.

Nomenclature

ANN	artificial neural network
COP	coefficient of performance
cGAN	conditional generative adversarial network
DH	district heating
GSHP	ground source heat pump
LSTM	long short-term memory
MAE	mean absolute error
MARS	multivariate adaptive regression splines
MPC	model predictive control
PCM	phase change material
RMSE	root mean square error
SM	surrogate model
SVR	support vector regression
TES	thermal energy storage

1. Introduction

Energy flexibility in district heating (DH) systems has gained significant attention due to its potential to optimize the use of renewable energy, waste heat, and thermal energy storage (TES) [1,2]. The flexibility is vital for enhancing system capacity, reducing operational costs, and is key to increasing overall efficiency and resilience, making the system more adaptable to varying energy demand and supply conditions [3]. In addition to traditional simulation tools, data-driven surrogate models (SMs) have become increasingly popular, enabling design optimization and operational improvements [4,5]. However, the

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effectiveness of SMs is strongly influenced by the quality and robustness of the data used in the model development. Therefore, it is essential to prioritize the development of reliable SMs and advance energy flexibility within DH systems.

DH systems are centralized infrastructures that supply heat to multiple buildings via a network of insulated pipes [6]. These systems play a vital role in reducing carbon emissions by efficiently utilizing heat from various sources, such as solar thermal, biomass, and waste heat, in combination with thermal energy storage [7–9]. This hybrid approach not only enhances flexibility but also allows for better management of intermittent energy supplies [10]. One of the key components in these systems is TES, which allows surplus heat to be stored and used during periods of high demand or when renewable energy supply is low [11–13], addressing both short-term and long-term (seasonal) flexibility needs [14,15]. Short-term solutions typically involve the use of sensible and latent heat storage [16]. Olsthoorn et al. [17] reviewed various modeling methods, focusing on the integration of different TES systems and renewable energy sources into DH networks to enhance building sustainability. Li et al. [18] demonstrated that incorporating a water tank into DH networks can effectively utilize waste heat to address supply-demand mismatch in the non-heating season and reduce peak demand during the heating season. Another study [19] examined the integration of two TES systems, comprising a short-term water tank and a long-term borehole system. The implementation of the water tank led to a 31 % reduction in peak load and a 5 % decrease in energy costs, while the borehole system enhanced waste heat recovery from 77 % to 96 % and contributed to an 8 % annual reduction in CO_2 emissions. These findings underscored the importance of combining demand-side and supply-side flexibility with water-based TES to reduce total heating demand and related operational costs. However, water has a relatively low energy density compared to other thermal storage mediums, such as phase change materials (PCMs). Consequently, for an equivalent amount of energy storage, water requires a significantly larger volume than higher energy density alternatives.

In addition to water-based TES integrated with DH systems, PCM storage has attracted considerable attention due to its high heat storage capacity [11]. For example, Wu et al. [20] proposed a design methodology for a heat accumulator aimed at peak shaving of heat sources, considering user heat loads and historical heating temperatures. It was found that phase-change temperature showed a significant impact on thermal efficiency, and the implementation of the accumulator resulted in a 63.21 % reduction in the discarded heat at the heating substation. Pans and Eames [21] applied a model to simulate a DH network in the UK, by incorporating short-term water-based and PCM-based storage, as well as a seasonal TES system. The results demonstrated that utilizing these storage solutions could reduce CO_2 emissions by 83.4 % and electricity costs by 12.3 % compared to a scenario without using thermal energy storage. Hlimi et al. [22] introduced a numerical approach using the apparent heat capacity method to evaluate the thermal performance of a storage tank filled with PCM-encapsulated spheres, which showed promise for designing latent heat storage systems in urban DH networks. The research emphasized the importance of selecting an appropriate PCM and identified an optimal melting temperature for achieving efficient hot water production. These studies demonstrated the benefits of PCM storage in improving efficiency and reducing emissions and costs in DH systems.

Regarding the design and optimization of an observed system, SMs, also known as metamodeling, reduced-order modeling, or proxy models, are designed to replace high-fidelity models [23]. These models have become essential in rapid design optimization, offering a computationally efficient alternative to traditional high-fidelity approaches [24]. They are particularly valuable in scenarios where fast decision-making is critical, as they can significantly reduce computational time without sacrificing accuracy [25]. For example, Sharifa and Hammad [26] employed an SM based on an artificial neural network (ANN) to analyze the data generated from a simulation-based multi-objective optimization model. The proposed model can estimate total energy consumption, life cycle costs, and life cycle assessment outcomes across different renovation strategies, offering a substantial reduction in computational time compared to traditional building energy simulations, while preserving a satisfactory level of accuracy. Similarly, Abokersh et al. [27] applied SMs to optimize the design of solar-assisted DH systems for different community sizes (10, 25, 50, and 100 buildings) in Madrid, Spain. The results demonstrated significant economic and environmental benefits, achieving a solar fraction above 82.1 % and an efficiency of over 69.5 %, particularly with the inclusion of seasonal storage tanks. Xu et al. [28] introduced a comprehensive SM combining an ANN with a conditional generative adversarial network (cGAN). When integrated with the non-dominated sorting genetic algorithm II (NSGA-II) optimization algorithm, this SM enabled rapid optimization of key parameters, improving computational efficiency by 99.97 %. Ren et al. [29] proposed a data-driven surrogate optimization approach for optimal deployment of multi-energy storage systems at a cluster level, aiming to minimize energy bills for building clusters under demand response schemes. In Ref. [30], based on the TRNSYS simulation data, a proxy model was developed using an ANN toolbox to calculate cost-optimal solar collector areas under various scenarios. In Ref. [31], linear regression, random forests, and support vector regressions were used to generate metamodels for quick and accurate annual building load simulation in a 5th generation DH and cooling system. While typical simulations took around 10 min, the proposed metamodels reduced runtime to less than 10 s, allowing for rapid simulations with user-defined covariates. Likewise, Thrampoulidis et al. [32] presented an SM designed to evaluate necessary building envelope and energy system measures for retrofits, using ANN to balance accuracy and computational costs. This model required a smaller input set and dramatically reduced the time needed to derive retrofit solutions from 3.5 min to 16.4 μs while maintaining high accuracy. These findings clearly demonstrated the benefits of SMs in design optimization, significantly reducing optimization time while maintaining high accuracy.

Beyond enabling fast design optimization through SMs, some researchers explored prediction and operational control strategies, which can also offer valuable benefits through these models. Quick et al. [33], for instance, utilized an SM to project the dynamics of the long-term electricity market. Similarly, Liang et al. [34] employed a surrogate modeling method to estimate high-resolution hourly cooling and heating demands throughout the year during the early design phase of buildings, which was essential for optimal system configuration and operation. In the context of energy demand prediction, Melo et al. [35] tested and compared various statistical modeling techniques, including multiple linear regression, multivariate adaptive regression splines (MARS), support vector regression (SVR), Gaussian process models, random forests algorithms, and ANN. The results showed that the ANN performed best for a medium-sized office, achieving a normalized root mean square error (RMSE) below 1 %.

On the operational strategies side, Wang et al. [36] presented an improved numerical model for a ground source heat pump (GSHP) system with variable input temperatures, alongside an XGBoost-based SM to accelerate the optimization. The optimized strategy reduced total operational time by 2–9 % through scheduling adjustments, highlighting substantial practical potential. Additionally, SMs can be effectively integrated into model predictive control (MPC) strategies to enhance operational optimization. For example, in Ref. [37], a long short-term memory (LSTM) SM, trained using EnergyPlus simulation data, was incorporated into an MPC framework, where a genetic algorithm optimized the energy-efficient operation of a space heating system. This approach reduced energy use by 7 % and energy costs by 13 %, with the potential for further savings. Zhang et al. [38] combined a simplified physical model with a data-driven surrogate approach to enhance the operational optimization of a DH substation. The physical model captured key operational behaviours and generated candidate control solutions, while the data-driven SM refined these solutions using historical data. This hybrid MPC strategy demonstrated strong operational efficiency. In addition, Pinto et al. [5] combined a data-driven LSTM SM with deep reinforcement learning to manage heat pumps and water storage for four buildings. This strategy led to a 3 % decrease in electricity costs and a 23 % reduction in peak energy demand without compromising indoor temperature control. The results from the aforementioned studies indicated that employing data-driven SMs for design optimization, predictive tasks, and operational optimization can deliver substantial benefits and enhanced efficiency. However, the critical role of datasets in strengthening the robustness and stability of these models is often overlooked in most control optimization studies.

Based on the above analysis, it is evident that TES can enhance the energy flexibility of hybrid DH systems, resulting in reduced operational costs and maximized system capacity. SMs are essential in enabling rapid design optimization, accurate prediction, and improved operational strategies across various energy systems. However, many studies primarily focused on using TES for peak shaving or balancing energy supply and demand, without fully addressing both peak shaving and load shifting simultaneously. From the perspective of heating price models [39], reducing operational costs requires a more integrated approach that considers both the variability of spot prices and the occurrence of peak demand in relation to load shifting and peak shaving. Therefore, a research gap remains in exploring effective strategies for shaping the energy demand profile through the integration of load shifting and peak shaving using a PCM-based thermal storage tank.

Another notable gap is that most studies emphasized using collected data to develop data-driven SMs for specific purposes or to improve the algorithms used to train these models, yet they often overlook the importance of diverse datasets in impacting the robustness and stability of these models. While developing SMs based on historical data from real-world systems is an effective way to improve performance, current system operations often rely on single, basic strategies, which can lead to limited data variability. Conventional control strategies make decisions based on system observations, such as building energy demand, meaning that control decisions are strongly influenced by the state of the system. As a result, similar system conditions lead to similar control decisions, reducing the opportunity to explore alternative decisions under comparable operational scenarios. The lack of diverse data can significantly affect the accuracy and robustness of SMs. Therefore, exploring the variability of simulation data through operational strategies for SM development represents another significant gap.

Consequently, this study aims to develop a robust SM based on simulation data to improve the energy flexibility of an integrated DH system with a PCM tank. The focus is to optimize both load shifting and peak shaving by utilizing the PCM tank guided by the robust SM. The key novelties of this study are as follows.

1) To address the complexity and high computational demand of traditional physical models, data-driven SMs leveraging machine learning techniques are developed. To investigate how different datasets affect the performance of these SMs, multiple SMs are constructed based on various datasets generated by implementing different demand-response operational strategies for managing the PCM tank within a hybrid DH system. The system was simulated using TRNSYS to represent real-world conditions.
2) To strengthen the energy flexibility of a DH system integrated with a TES, an optimized operational strategy is introduced. Given the heating price model, the strategy for managing the tank, incorporating peak shaving and load shifting, is proposed and implemented in the hybrid DH system using the best-performing SM. The strategy is then applied in a TRNSYS environment for further evaluation.

This study demonstrates how to leverage simulation data, generated by suitable operational strategies, to develop a robust SM aimed at enhancing the energy flexibility of DH systems.

2. Development of data-driven surrogate models and operational strategies

2.1. Overview of the overall methodology

Fig. 1 illustrates the proposed methodology, comprising three main steps: 1) data generation under various basic operational strategies; 2) developing and training SMs and evaluating their robustness; and 3) developing an optimal operational strategy that can be used for both peak shaving and load shifting.

In the first step, data is generated from a simulated DH system integrated with a data centre and a PCM tank under various operational strategies. To investigate how different operational strategies influence historical data generation and SM development, various demand-response strategies are implemented in order to manage the PCM tank. These strategies include a single setpoint price-based strategy, a multi-setpoint demand-based strategy, a daily time-based strategy, and a day-ahead price-based strategy. An integrated energy system was simulated using the TRNSYS platform, which generates data for each strategy, including PCM average temperature, waste heat temperature, heat pump electricity use, solid fraction, heat supply, and the return water temperature from the data centre. These datasets are then prepared for the subsequent step.

In the second step, data-driven SMs are developed to overcome the complexity of traditional physical models and improve system performance. A key focus is to evaluate the robustness of the models and examine how datasets generated under different operational strategies influence the performance of the models. LSTM networks are employed to develop the SMs using each dataset, with the key parameters and input/output variables identified based on domain expertise. Model robustness is further assessed by introducing an independent dataset from a TRNSYS-based random strategy. Its performance is assessed by comparing the surrogate model predictions with the test data using mean absolute error (MAE) and RMSE metrics. The accuracy of the best-performing SM is validated by comparing its simulated results to the data obtained from TRNSYS.

Finally, an optimal operational strategy that combines moving average control and setpoint control is proposed to manage the PCM tank, aiming to facilitate peak shaving and load shifting to improve the energy flexibility of the integrated DH system. This strategy, based on the best-performing SM, is optimized using grid search and then applied to the TRNSYS simulation platform. The effectiveness of the strategy is evaluated in terms of three key aspects, including total operational costs, peak demand, and price-responsive heating costs.

2.2. Data generation and collection

Fig. 2 depicts a schematic of a DH system integrated with a data centre and a PCM tank. In this setup, waste heat recovered from the data centre is utilized to preheat return water from the buildings, reducing the overall heat use of the system. The PCM tank plays a key role in managing the heating demand profile and enhancing the energy flexibility of the hybrid DH system. The PCM tank plays a key role in managing the heating demand profile and enhancing the energy flexibility of the hybrid DH system. It was chosen instead of conventional water-based storage because its superior thermophysical properties allow for greater system flexibility. Compared to water-based storage, PCMs exhibit a much higher volumetric latent heat (~368 kJ/m^3), which allows greater energy density and a significantly reduced storage footprint. For instance, storing 10^6 J requires about 16,000 kg and 16 m^3 of water, whereas only 4350 kg and 2.7 m^3 of inorganic PCM are needed [40]. Economically, PCM systems can achieve a payback period of ~8.7 years [41], while pit water storage systems are reported at 100–300 USD/kWh [42] depending on scale and configuration.

The main DH substation includes two heat exchangers. One is used to heat the preheated water to the desired temperature for distribution to

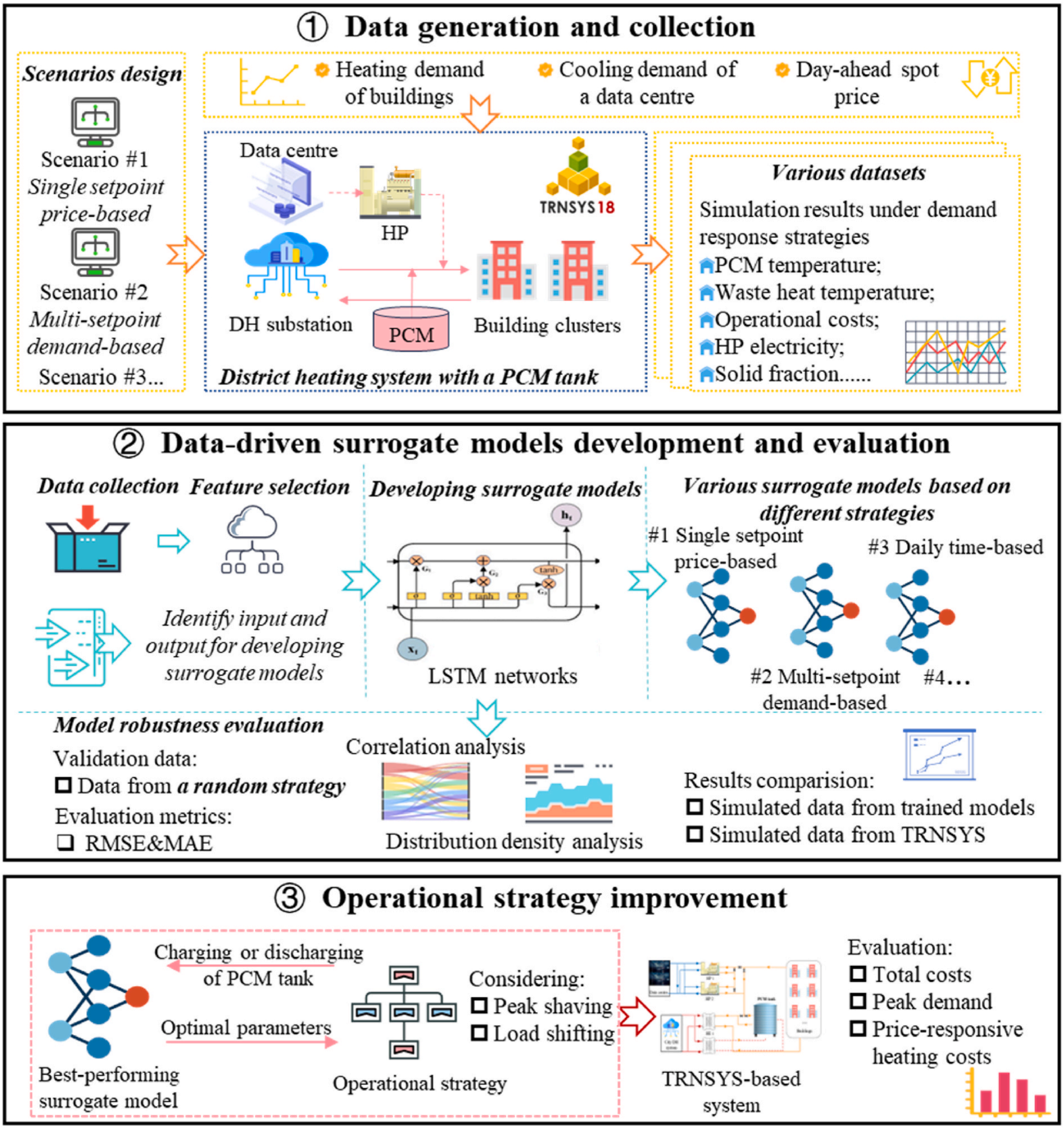


Fig. 1. Outline of the overall methodology.

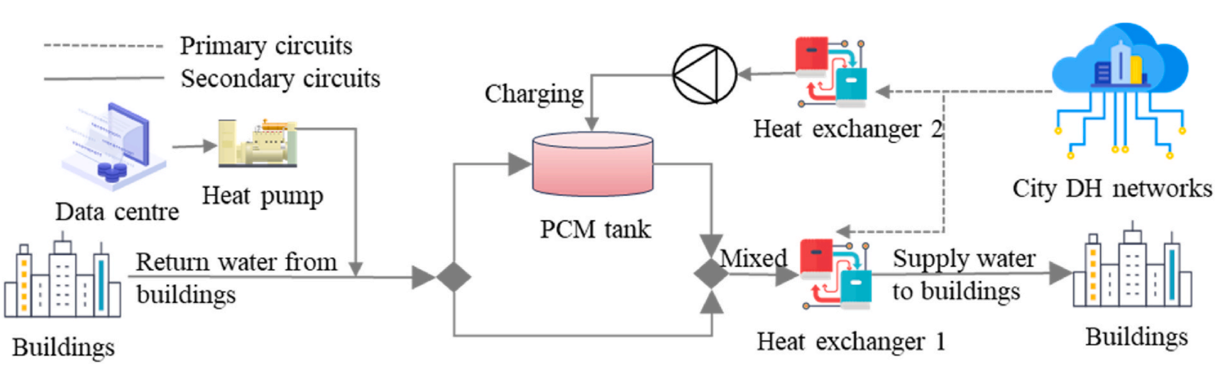


Fig. 2. Configuration of a DH system with a data centre and a PCM tank.

the buildings, and the other is dedicated to charging the PCM tank according to the demand response strategies. This configuration enables optimized energy usage and improves overall system flexibility.

Operational strategies are essential for effective management of an integrated DH system, particularly in optimizing the performance of a PCM tank. The operation of a PCM tank can significantly influence both the overall system performance and operational costs. Data-driven SMs are gaining popularity because they simplify the development of traditional physical models for complex systems and enable rapid modelling of building energy performance. In other words, instead of using physical models to build and validate system performance, data from operational systems can be collected to construct models to improve the overall performance. However, the effectiveness of these SMs primarily depends on the quality, diversity, and accuracy of the data used for training. In real-world energy systems, operators often rely on simple, basic strategies to ensure the safety and reliability of hybrid DH systems. While in practice, these strategies tend to produce data that lacks the necessary diversity required to train robust SMs. Therefore, the choice of operational strategies is critical, as it directly influences the variance and richness of the data generated, ultimately affecting the reliability and effectiveness of data-driven SMs.

For system operations, multiple strategies can be employed to manage the operation of a PCM tank within hybrid DH systems. The schematics of the four main operational strategies are shown in Fig. 3 and described as follows:

Single setpoint price-based control: The spot price is used as an external signal to manage the charging and discharging of the PCM tank to improve energy flexibility and reduce heating operational costs by optimizing energy usage throughout the heating season. The control strategy operates based on the following logic: when the current spot price is lower than a predetermined setpoint, indicating a valley price period, the PCM tank is then charged via the heat exchanger until it reaches its temperature limit. Conversely, when the current spot price exceeds the setpoint, indicating a peak price period, the PCM tank is discharged to serve as a heat source to supply heat to buildings.

Multi-setpoint demand-based control: This strategy takes into account two key factors, including time (seasonal periods) and setpoints. Due to the significant differences in heating demand during the mid-season compared to the early and late seasons, the heating season can be divided into three periods: early, mid, and late. During the early and late periods, when the heating demand exceeds the setpoint a_1, see Fig. 3, indicating peak demand, the PCM tank discharges heat to shave the peak demand. Otherwise, the heat exchanger charges heat into the PCM tank. During the mid-period, the control logic is similar but requires an additional setpoint a_2 for finer control. Typically, a_1 is set lower than a_2.

Daily time-based control: This strategy considers the daily variation in heating demand for buildings. For example, within a typical day, heating demand is the highest between 6.30 a.m. and 3.30 p.m. During this period, the PCM tank can be discharged to shave the peak. During the remaining hours, the PCM tank is charged using a heat exchanger until it reaches the desired temperature.

Day-ahead price-based control: In this strategy, the spot price serves as an external signal to manage the charging and discharging of the PCM tank. Since the spot price is determined day-ahead, the PCM tank discharges heat during peak pricing when the current price surpasses the average 24-h price for the day. Conversely, when the current price is lower than the average 24-h price, the PCM tank is charged using a heat exchanger until it reaches the desired temperature.

In this study, these operational strategies were implemented into a DH system with a data centre and a PCM tank, and simulated using the TRNSYS platform.

The building heating demand and the cooling demand of the data centre were the actual measured data obtained from a case study to be presented in Section 3. The models for the PCM tank, water pumps, and controllers used were sourced from the standard TRNSYS and TESS component libraries. For example, Type 9e was used to import time-series data for the heating demand of buildings, data centre cooling load, and spot prices. Type 682 simulated ideal air-conditioning system performance, while Type 225 was a water-to-water heat pump. The PCM-based TES was modelled using Type 1334, and Type 114 was employed to simulate single-speed pump operation. Data was generated and collected for each strategy, and the performance was then evaluated.

2.3. Surrogate model development and evaluation

2.3.1. Surrogate model development

Data-driven SMs offer significant advantages, particularly in complex energy management systems [43]. A key benefit is that such models can drastically reduce computational time compared to traditional physics-based models, enabling faster optimization, prediction, and

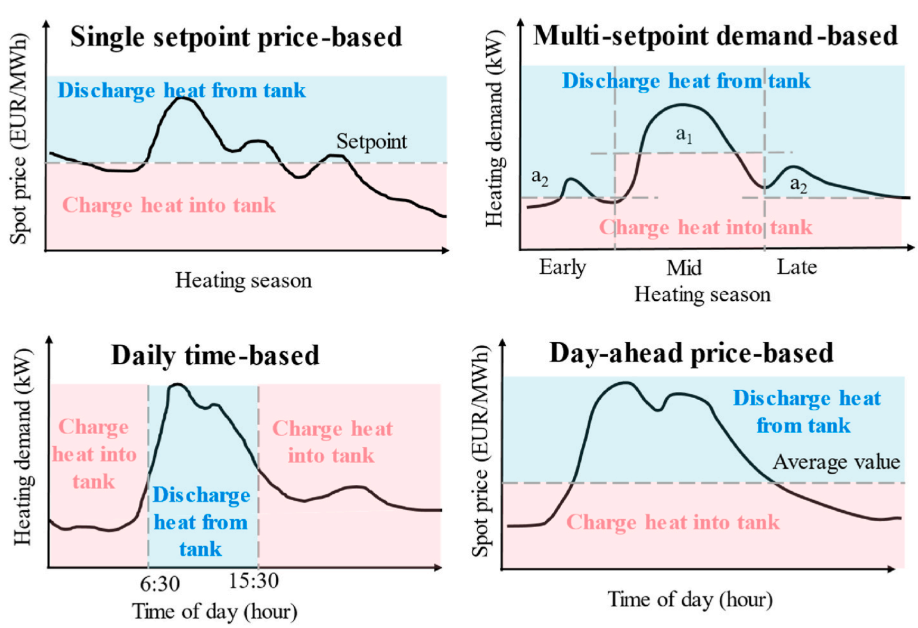


Fig. 3. Schematics of the four operational strategies.

control. By leveraging historical data to approximate system behaviour, such models can simplify the modelling process and eliminate the need for detailed physical models [44]. In other words, with sufficient historical data, SM models can be developed directly to optimize operational strategies and enhance system performance. This approach eliminates the traditional requirement of first constructing detailed physical models before developing surrogate models.

Two of the most critical aspects of developing SMs are to identify the inputs and outputs of the target models and select an appropriate training algorithm. In this study, the inputs and outputs of the SM were determined by analyzing the operational characteristics and performance metrics of the DH system with a data centre [45]. The inputs of the model included the average temperature of the tank, solid fraction of the tank, coefficient of performance (COP) of heat pumps, heat pump electricity use, the waste heat temperature from the heat pumps, the inlet and outlet water temperatures of the tank, heating demand of buildings, cooling demand of the data centre, and charging or discharging actions of the tank, at the time step t. The outputs of the model included the average temperature of the tank, solid fraction of the tank, COP of heat pumps, heat pump electricity use, the waste heat temperature from heat pumps, and the inlet and outlet water temperatures of the tank at the time step $t+1$.

LSTM networks were used to develop the SMs, due to their effectiveness in time-series prediction and sequence modeling tasks [46–48]. The configuration of the LSTM model is shown in Fig. 4. The model consists of a series of interconnected memory cells, each employing three gates, including the input gate, the output gate, and the forget gate. These gates manage the flow of information through the network. The forget gate decides which parts of the previous data should be retained or removed. The input gate handles the incorporation of new data from the current time step, and the output gate forwards the processed information to subsequent steps. These gates facilitate the flow of information through specialized, self-connected units, preventing gradient vanishing and enabling the model to capture long-term dependencies effectively.

The forget gate is determined using Eq. (1). The input gate is determined using Eqs. (2) and (3) [49].

$$f_t = \sigma\left(W_f \cdot [h_{t-1}, X_t] + b_f\right) \tag{1}$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, X_t] + b_i) \tag{2}$$

$$\widetilde{C}_t = tanh(W_C \cdot [h_{t-1}, X_t] + b_C) \tag{3}$$

where f_t is the output of the forget gate, W_f, W_i, and W_C are the weight matrices, h_{t-1} is the hidden state from the previous time step, X_t is the input at the current time step, b_f, b_i, and b_C are the bias terms, σ is the sigmoid activation function, i_t is the output of the input gate, and $\widetilde{C}_t$ is the candidate cell state.

The memory cell state is updated by combining the forget gate and the input gate based on Eq. (4) [49]. The output gate is determined using Eqs. (5) and (6) [49].

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \widetilde{C}_t \tag{4}$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, X_t] + b_o) \tag{5}$$

$$h_t = o_t \cdot tanh(C_t) \tag{6}$$

where C_t is the updated cell state, C_{t-1} is the previous cell state, o_t is the output of the output gate, h_t is the hidden state at time step t, W_o is the weight matrix for the output gate, and b_o is the bias term.

In this study, the hyperparameters were set as shown in Table 1: the learning rate was 0.01, and the batch size was 128 based on trial-and-error testing. The SMs were developed using Python 3.10.

2.3.2. Model robustness evaluation

Evaluating model robustness is a critical step in validating SMs. In this study, the validation dataset was generated using a randomized control strategy. This strategy employed Type 577 in TRNSYS to produce control signals (0 or 1), indicating whether heat should be charged into or discharged from the tank. In the component settings, the initial random seed was set to 60. By introducing randomness into the control process, the robustness of the SM was assessed under varying conditions.

Various measures are available to evaluate the accuracy of SMs in predicting values or classes that closely match the observed data. In this study, given that the data include zero values, two commonly used metrics, MAE and RMSE, were employed.

MAE measures the average absolute difference between the predicted and actual values [50]. It gives a straightforward average error magnitude, treating all errors equally. MAE is easy to interpret, with units consistent with the original data.

Table 1
Hyperparameter settings for the SMs.

Item	Value
Number of LSTM layers	1
Number of Dense layers	1
Units in LSTM layer	50
Units in Dense layer	13
Activation function	ReLU (used in both LSTM and Dense layers)
Optimizer	Adam
Learning rate	0.01
Loss function	Mean Squared Error
Maximum epochs	500
Batch size	128

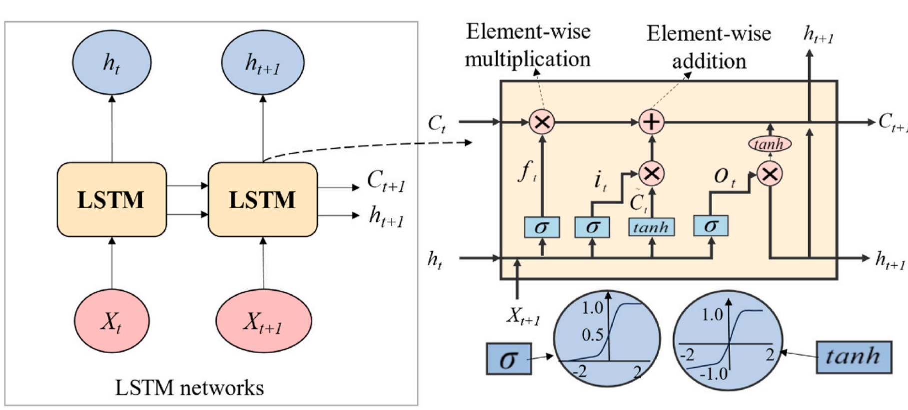


Fig. 4. Illustration of the LSTM model (Note: the definitions of the symbols used are provided in the following equations).

$$MAE = \frac{1}{n} \times \sum_{i=1}^{n} |y_i \text{-} \hat{y}_i| \tag{7}$$

RMSE calculates the square root of the average squared differences between predicted and actual values [50]. It emphasizes larger errors by squaring them, making it more sensitive to outliers.

$$RMSE = \sqrt{\frac{1}{n} \times \sum_{i=1}^{n} |y_i - \hat{y}_i|^2} \tag{8}$$

where n is the number of data points (set to 5088), y_i is the actual value for the ith data point (from TRNSYS), and $\hat{y}_i$ is the predicted value for the ith data point (from SMs).

2.4. Improving energy flexibility in hybrid DH systems

2.4.1. DH and electricity price model

A comprehensive analysis of generalized models for heating and electricity pricing is presented in Ref. [51]. The heating price model mainly emphasizes heating use and peak demand charges, as these elements are commonly included in current pricing mechanisms. In several European countries, electricity pricing typically includes grid fees, taxation, and a variable rate determined by market conditions [52]. In some cases, an additional fee is applied for excessive peak loads. As this study observed a much higher heating demand compared to electricity consumption, the peak load surcharge related to electricity was omitted. The overall monthly system cost is calculated by summing the expenses for both heating and electricity.

2.4.2. Proposed operational strategy for improved energy flexibility

In this study, heating costs accounted for the largest proportion of total operational costs. To reduce these costs, it is essential to lower both the heating energy demand and peak demand costs, as identified through the analysis of the heating price model. These costs are influenced by the spot price and peak load. Therefore, effectively managing peak shaving and load shifting is crucial for optimizing complex energy management.

The control logic, illustrated in Fig. 5, aims to improve energy flexibility by considering both peaking shaving and load shifting. The strategy prioritizes peak shaving based on building heating demand first and then considers load shifting based on the spot price. The control logic for managing the PCM tank is as follows.

- When the heating demand of buildings exceeds a setpoint a, the PCM tank discharges heat to shave the peak demand.
- If the heating demand is below the setpoint and the current price is higher than the future 24-h average price plus a constant c, the PCM tank also discharges heat.
- If the tank is not in a discharging state, and the current price is lower than the future 24-h average price plus a constant g, the PCM tank is charged. In all other cases, the PCM tank remains idle, neither charging nor discharging.

This strategy was tested using two months of data, and the parameters a, c, and g were optimized through grid search. Grid search was selected due to its simplicity and its ability to systematically evaluate all possible combinations within the predefined ranges. For each parameter combination, the SM was used to predict system performance, and the corresponding total operational cost was calculated. The parameter set that yielded the lowest cost was identified as the optimal strategy. After optimization, the strategy was implemented in the TRNSYS simulation system, where its performance was evaluated based on the following indicators.

The strategy will be evaluated based on three key aspects: price-responsive heating costs, peak demand, and total operational costs.

Price-responsive heating costs **($C_{\text{load\_}mon}$)**: This indicator measures the heating cost component that responds to fluctuations in the spot price and reflects the economic manifestation of load shifting. In principle, load shifting refers to moving heating demand from high-price periods to low-price periods. Under the proposed strategies examined in this case study, however, it is difficult to quantify the absolute shifted load in terms of thermal energy, because multiple effects (e.g., peak shaving, waste heat utilization) may also contribute to variations in heat supply. To address this, a cost-based indicator that captures the extent to which heating demand is exposed to spot price fluctuations is proposed.

The indicator is expressed as the heating energy supplied at each time step multiplied by the corresponding spot price (plus tax surcharge), aggregated over the evaluation period, as shown in Eq. (9). A lower value of C_{load_mon} indicates that the heating demand was successfully shifted away from high-price periods, achieving greater price-responsive load shifting. In this sense, the indicator translates the concept of load shifting into an economic dimension that can be directly compared across operational strategies (e.g., daily time-based vs. proposed strategy in this study).

$$C_{\text{load\_}mon} = \sum_{i=1}^{n} HE_i \cdot \dot{Q}_i \cdot \Delta t \tag{9}$$

where HE_i refers to the heating energy demand price, calculated as the spot price from the power market plus a tax surcharge of 0.2294 EUR/MWh, as provided by the local DH company (Statkraft Varme, Trondheim [53]), and $\dot{Q}_i$ refers to the heating rate supplied to users.

Peak demand **($\dot{Q}_{sur}$)**: This indicator evaluates the extent to which peak demand can be reduced, comparing the proposed strategy to the baseline. It reflects the effectiveness of the strategy in managing peak demand, as shown in Eq. (10).

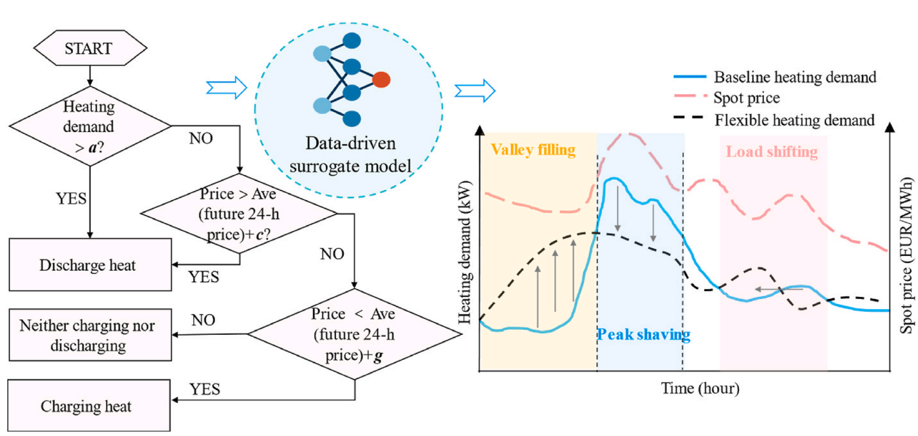


Fig. 5. Flexible heating demand management with the new strategy.

$$\dot{Q}_{sur} = \frac{\dot{Q}(1) + \dot{Q}(2) + \dot{Q}(3) + \ldots + \dot{Q}(n)}{n} \tag{10}$$

where $\dot{Q}(1)$, $\dot{Q}(2)$, $\dot{Q}(3)$, $\cdots$, $\dot{Q}(n)$ represent the hourly heating demands sorted in descending order ($n = 3$ in this study) [54]. Peak demand was defined as the average of the three highest hourly heating demands during the evaluation period, providing a more robust representation of peak conditions and reducing the influence of isolated short-term extremes.

Total operational costs **(C_{costs_mon}):** This indicator reflects the overall costs associated with the system use of heating and electricity, as shown in Eq. (11) [51].

$$C_{costs\_mon} = C_{load\_mon} + f_{sur} \cdot \dot{Q}_{sur} + 1.25 \cdot \sum_{i=1}^{n} \dot{E}_i \cdot e_i \cdot \Delta t + f_{ren} \cdot E_i + \frac{F}{12} \tag{11}$$

where f_{sur} refers to the tariff applied to peak heating loads (i.e., 3.8 EUR/kW/month), f_{ren} refers to a grid rent fee set at 0.023 EUR/kWh, $\dot{E}_i$ refers to the electricity use rate, E_i is the integral electricity demand, e_i is the spot price from the power market, and F represents the fixed fee of 190 EUR/year [52,55]. In this case study, electricity accounted for less than 10 % of the total operational costs. Therefore, the proposed strategy primarily focused on reducing heating-related costs. If electricity demand was to represent a larger share, the strategy would need to be extended to incorporate electricity flexibility.

3. Description of the case study

3.1. Case study description

The methodology was evaluated using a DH system that incorporates a data centre, located on a university campus in Norway. Fig. 6 illustrates the configuration of the DH network, including the integration of data centre waste heat. This system delivers heating to a range of campus buildings, collectively spanning about 300,000 m^2. Additional information on these buildings can be found in earlier studies [19,56]. Throughout the year, the data centre supplied approximately 20 % of the total heating demand, with the remaining 80 % provided by the urban DH system via the main substation. It should be emphasized that the analysis in this study pertains specifically to a campus-scale DH configuration.

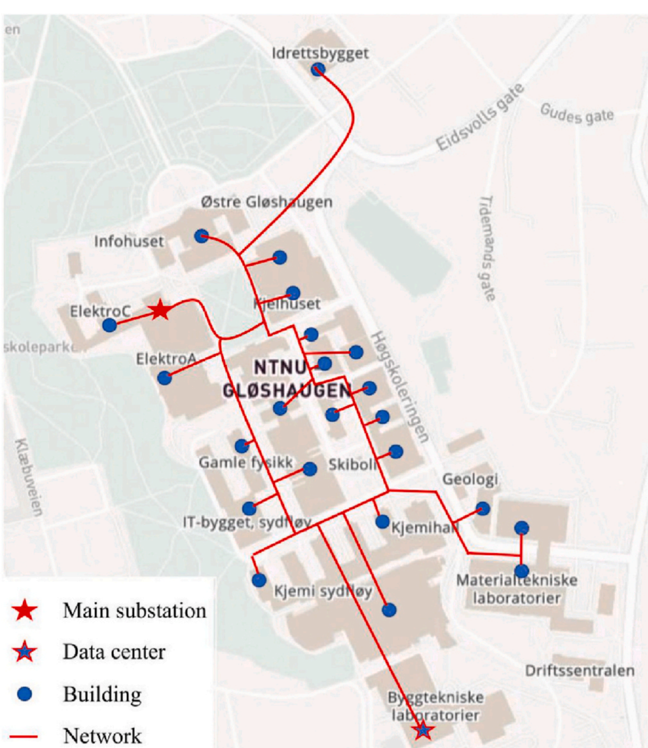


Fig. 6. Configuration of the campus DH network integrated with a data centre [19].

The schematic of the analyzed hybrid DH system is illustrated in Fig. 7, which consists of a data centre, two series-connected heat pumps, two water pumps, a main DH substation equipped with two heat exchangers, a PCM tank, and a campus DH network. In this study, the main substation supplied the primary heating energy for the buildings. Waste heat from the data centre was used to preheat the return water from the buildings, which then entered one of the two heat exchangers to reach the desired supply temperature. The second heat exchanger facilitated heat charging into the PCM tank, enabling load profile management in response to external control signals.

For the TRNSYS simulation, an input file was required, which included day-ahead spot prices, the data centre cooling demand, and the heating demand of campus buildings. Hourly data covering the heating season from October 2017 to April 2018 were used in the analysis. As illustrated in Fig. 8, the cooling energy provided by the data centre remained relatively consistent, while heating demand and spot market prices exhibited significant variability over the observed period.

The properties of the PCM used for the storage tank are listed in Table 2. With a phase change temperature of 58 °C, these materials were well-suited for application in thermal management within DH systems. Further details can be found in Ref. [57]. In this study, the size of the PCM tank was set to 120 m^3 based on prior research [18,51,58] and the system's heating demand capacity. Although previous studies focused on large-scale water tanks, their findings were adapted by accounting for the thermal storage capacity of the PCM to determine an appropriate PCM tank size. The heat loss coefficient was 0.1 kJ/h.m^2.K. The flow rate was set at 88,000 kg/h, and the maximum allowable temperature was limited to 70 °C.

Another essential component of the hybrid DH system is the set of heat pumps and water pumps. The nominal performance characteristics of the equipment are detailed in Refs. [51,59].

3.2. Scenarios development

To investigate the impact of the operational strategies on the variance of data and the robustness of SMs, various operational strategies were implemented into the DH system with a data centre and a PCM tank. These strategies focused on how to manage the charging and discharging of the PCM tank.

- The baseline scenario was a DH system operating with a data centre without thermal energy storage.
- *Scenario #1 (single setpoint price-based control):* Utilized spot prices as an external signal to control the PCM charging and discharging. The setpoint used was 36 EUR/MWh, which was determined based on the

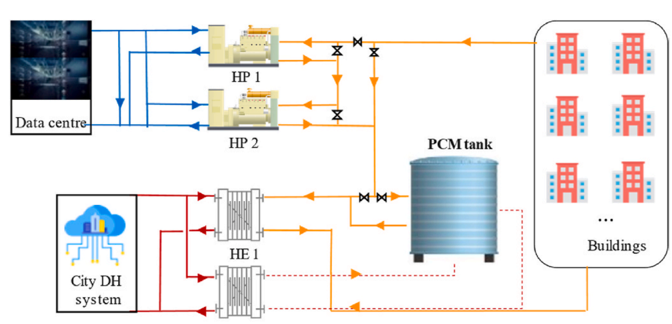


Fig. 7. Schematic of the integrated DH network with a PCM tank.

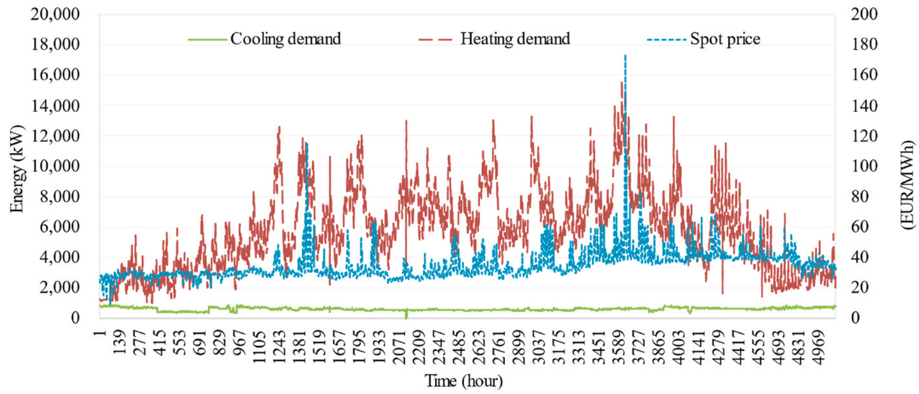


Fig. 8. Monitored energy usage and day-ahead spot prices.

Table 2
Physical properties of sodium acetate trihydrate and 10 % graphite.

Physical properties	Value
Density in solid state	1360 kg/m^3
Density in liquid state	1260 kg/m^3
Kinematic viscosity	5 m^2/s
Liquid heat capacity	2505 J/kg ^{o}C
Solid heat capacity	1900 J/kg. ^{o}C
Expansion coefficient in liquid phase	0.001
Solid conductivity	2.3 W/m ^{o}C
Liquid conductivity	2 W/m ^{o}C
Melting temperature	58 °C
Melting enthalpy	180–200 kJ/kg

trend and peak values of spot prices. Several values (e.g., 34, 36, 38, 40, and 42 EUR/MWh) were manually tested, and 36 EUR/MWh was found to offer the best performance.

- *Scenario #2 (multi-setpoint demand-based control)*: Divided the heating season into three periods, including early, mid, and late, and each with distinct setpoints ($a_1 = 6543$ kW for mid-season, $a_2 = 2218$ kW for early and late seasons) based on the building heating demand profiles. These setpoints and seasonal boundaries were determined by analyzing the seasonal variation in demand.
- *Scenario #3 (daily time-based control)*: Targeted daily peak demand by discharging the PCM tank between 6:30 a.m. and 3:30 p.m., with charging occurring outside these hours via a heat exchanger. Setpoints were derived from clustering results [45].
- *Scenario #4 (day-ahead price-based control):* Adjusted PCM charging and discharging based on spot price fluctuations, charging when prices were below the daily 24-h average. Discharging the PCM tank when the prices were above the daily 24-h average.

Each scenario generated datasets for training SMs, corresponding to scenario numbers #1 to #4. Table 3 summarizes these strategies.

Table 3
A summary of multiple operational scenarios.

No.	PCM tank	Operational strategy	Signal	Parameter settings
Baseline	No	No	No	No
1	Yes	Single setpoint price-based	Spot prices	36 EUR/MWh
2	Yes	Multi-setpoint demand-based	Time and heating demand	6543 kW and 2218 kW
3	Yes	Daily time-based	Time	6.30 a.m. to 3.30 p.m.
4	Yes	Day-ahead price-based	Spot prices	The daily 24-h average

4. Results and discussion

4.1. Impact of multiple operational strategies on operational costs

The aforementioned strategies were applied to the DH system integrated with a PCM tank, and simulated using the TRNSYS over a one-year heating season. An analysis of the simulation outputs, including the COP, variations in temperature and solid fraction within the PCM tank, supply temperature from the heat exchanger, and return temperature from users, confirmed the reliability of the results within the TRNSYS platform. Fig. 9 displays the total operational costs for the entire heating season for all the considered scenarios. The results indicated that compared to the baseline, the integration of a PCM tank reduced operational costs by optimizing the periods of energy usage. Among the tested strategies, Scenario #3 was the most effective one, achieving a 1.2 % reduction in the total operational costs relative to the baseline. However, the cost reduction was relatively minor, possibly due to the control setpoint not being fully optimized. Additionally, another contributing factor could be the influence of waste heat used for preheating the return water, which may reduce the heat exchange rate in the PCM tank and thereby limit its effectiveness.

4.2. Evaluation of various surrogate models

Five SMs were trained using LSTM networks based on three years of heating season data, covering the baseline and SMs for Scenarios #1 to #4. To evaluate their robustness, the validation dataset was created using a randomized control strategy to define the operational mode of the PCM tank. This approach employed a random module in TRNSYS to generate signals that dictated whether the tank should be charged or discharged. This approach was chosen because, in most cases, SM development aims to improve system performance by adjusting the parameters. Using a randomized dataset helps validate the robustness of models. Two indicators were used to assess the performance of the SMs: MAE and RMSE.

Generally, larger MAE and RMSE values indicate that a model is performing poorly, reflecting greater differences between the predicted and actual values. Fig. 10 presents the MAE and RMSE values of various SMs. From Fig. 10, it can be seen that the baseline SM exhibited the largest deviations in terms of both indicators compared to the other models. This is primarily because, in the baseline strategy, the system operated without a PCM tank, and the data used to train the baseline model did not include any actions such as charging or discharging. Consequently, the baseline model lacked the necessary information to accurately predict system performance, leading to higher errors in its prediction.

According to the evaluation indicators, SMs #1 and #2 performed better than the baseline model. This improvement was attributed to their training data, which included actions such as heat charging and

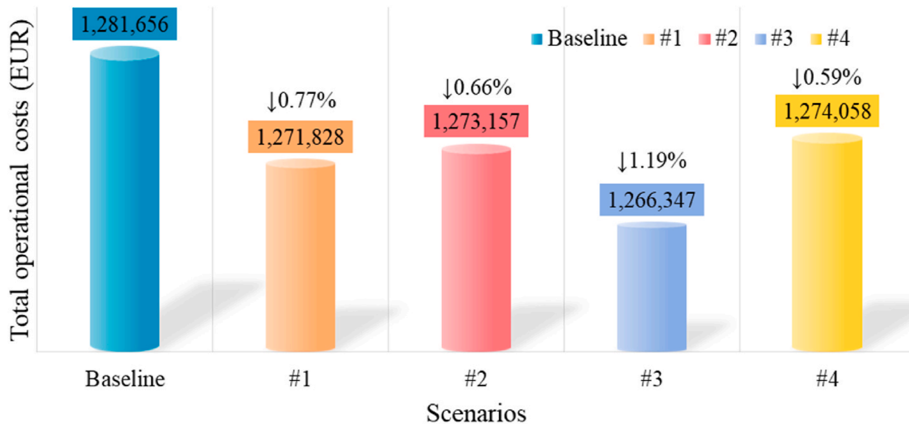


Fig. 9. Total operational costs under different strategies.

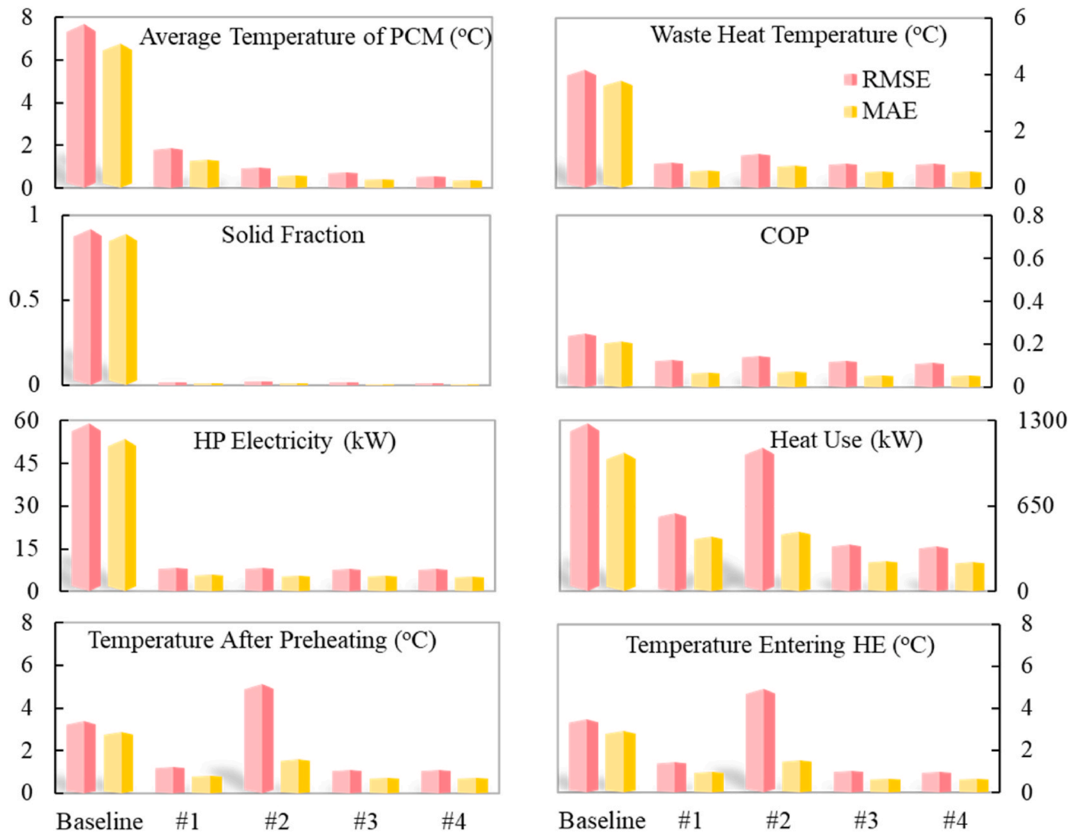


Fig. 10. Comparison of the baseline and SMs (#1-#4) in terms of RMSE and MAE performance metrics.

discharging in the PCM tank, introducing more variance and essential information in the dataset. As a result, these models were better equipped to predict system performance, leading to improved performance compared to the baseline model.

SMs #3 and #4, trained on the datasets generated by daily time-based and day-ahead price-based control strategies, respectively, demonstrated greater stability in terms of the MAE and RMSE values. Unlike SMs #1 and #2, SMs #3 and #4 could adjust actions within a single day, resulting in more diverse and richer datasets. Consequently, the training datasets for SMs #3 and #4 were more varied than those for the baseline model and SMs #1 and #2. This may explain why the values of the indicators were lower, leading to a more stable model. Moreover, SM #4 emerged as the best-performing model based on the comparison of the indicators.

In summary, the robustness of the SMs was largely attributed to the diversity of training data and the design of the control strategies. Diverse data exposure supported better generalization by allowing the model to learn from a broader range of information, thereby reducing the risk of overfitting. This aligned with the bias-variance trade-off theory, which suggested that balanced training data can minimize variance and enhance stability across different conditions.

Since different SMs employed the same neural network structures, the variation in their performance primarily stemmed from the differences in the training data used. The SM aims to capture the system state transitions under different actions. Specifically, at a given state, the system may transition to different states depending on the action taken. A well-trained SM can predict future system performance based on various control decisions. In this context, strategies that explore multiple actions under similar system states contributed to training a more accurate SM by enriching the dataset. Therefore, the datasets generated by different control strategies are analyzed to assess their effectiveness in exploring control decisions.

The system states are characterized by a wide range of variables, such as flow rate, temperature, and power. Among these variables, PCM tank temperature was selected as the representative variable to define the system state, as it directly impacts heating supply, energy flexibility, charging and discharging cycles, and interactions with other variables. Meanwhile, the main control actions at each time step were considered as the action variables for the analysis.

Fig. 11 illustrates the correlation between actions and average PCM tank temperatures, as well as the density distribution of different actions taken at various PCM tank temperatures. Action 1 represented the scenario in which a fraction of the return water from users entered the PCM tank, increasing its temperature. In contrast, action 0 represented the scenario where the return water from users bypassed the PCM tank and directly entered the heat exchangers to reach the desired temperature, while a separate heat exchanger was used to charge the PCM tank. The density curves of action 1 and action 0 indicated a similar likelihood of occurrence at a specific state when their density values were comparable. Otherwise, if one action had a higher density than the other, it was more dominant in occurrence.

To better evaluate the diversity in decision making under the same system states, an overlap area is introduced, which is the shared region between two probability density functions, as defined in Eq. (12).

$$\textit{Overlap area} = \int_{-\infty}^{\infty} \min\left(f_1(x), f_2(x)\right)dx \tag{12}$$

where $f_1(x)$ and $f_2(x)$ are the probability density functions of the two datasets, and $\min(f_1(x), f_2(x))$ is the minimum value of the two densities at each point x, representing the overlapping density.

A larger overlap area in the density curve indicates that actions are more evenly distributed across all possible temperatures. This is beneficial because exploring different actions under the same system state provides better insights into system performance. Specifically, in such a state, action 1 cannot be inferred from action 0, and vice versa, as the two actions were fundamentally opposite. Ensuring a well-distributed dataset with balanced actions enhanced the robustness of surrogate modelling.

From Fig. 11(a) and (b), it can be observed that datasets #1 and #2 exhibited a broad distribution, indicating that the basic control strategies were not effective in regulating the PCM tank temperature. Sometimes the tank temperature fluctuated within the range of 30–40 °C. Additionally, the number of action 1 instances significantly exceeded that of action 0, which may result in the trained model being more accurate in capturing the system performance under action 1 while the prediction of action 0 may not be accurate. Based on the overlap area, the numbers of action 1 and action 0 were not well-distributed. Combined with the analysis in Fig. 11, the SMs #1 and #2 demonstrated suboptimal performance.

In contrast, datasets #3 and #4 demonstrated a more balanced distribution, as the numbers of the two actions were more evenly distributed. Analysis of the two evaluation metrics (RMSE and MAE) further confirmed this observation. The overlap areas for datasets #3 and #4 exceeded 0.419, indicating that the actions in these datasets were well-distributed. Based on the value analysis and evaluation metrics, dataset #4 exhibited the best performance.

The overlap areas among different datasets are compared and summarized in Table 4.

Through evaluation and comparison, SM #4 emerged as the best-performing model. This SM was then used to simulate the system performance over a one-week period, which was compared with the simulation data obtained from TRNSYS. As shown in Fig. 12, the differences between the simulated data based on the SM and the data from the TRNSYS model were minimal, with only slight deviations. The predicted trends closely matched the TRNSYS-based data, indicating that SM #4 is robust and accurate.

The superior efficiency of SM #4, trained on the dataset exhibiting greater variance compared to the other models, suggested an important consideration, that is, greater data diversity may lead to a more robust SM. These findings could provide valuable insights for real-world systems. By collecting more diverse data through accurate operational strategies, data-driven SMs can become more stable and reliable.

4.3. Performance evaluation of the proposed operational strategy

To enhance energy flexibility and reduce total operational costs while considering heating price models, a new operational strategy was developed to manage the charging and discharging of the PCM tank in the DH system integrated with a data centre. This strategy prioritizes peak shaving to reduce peak building heating demand, followed by load shifting in response to varying spot prices. The control logic for managing the PCM tank has been presented in Section 2.4.2. This strategy was tested using the simulation data from a two-month period.

The new strategy was applied to the hybrid DH system based on SM #4. Grid search optimization was conducted to minimize the operational costs by testing parameter combinations: $a = [36,38,40,42,44,46]$ *221.8 (221.8 was a scaling factor), $c = [0, 1, 2, 3, 4]$, and $g = [-4, -3, -2, -1, 0]$. The optimal parameters identified were: a setpoint of 9316 kW for a, 2 EUR/MWh for c, and −2 EUR/MWh for g, as shown in Fig. 5. Subsequently, this strategy incorporating with optimized setpoints was applied to the TRNSYS simulation.

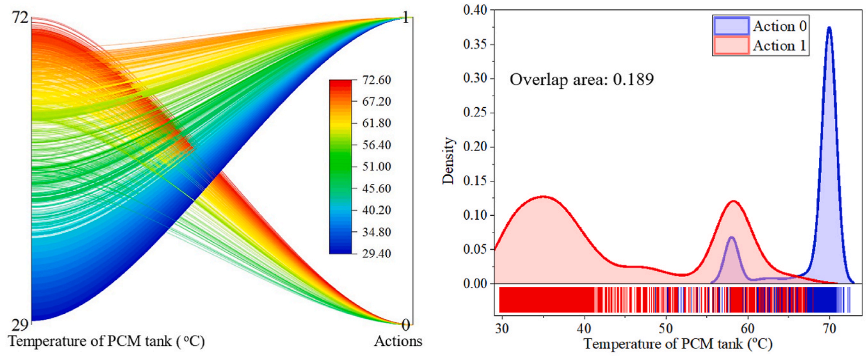
A short data segment was selected to demonstrate the effectiveness of flexible heating demand management, as illustrated in Fig. 13. The results showed that the strategy successfully achieved both peak shaving and load shifting. For example, when the heating demand exceeded a predefined threshold, the PCM tank discharged heat to reduce the peak. During periods of high spot prices, the system also discharged the PCM to avoid expensive heat usage. Conversely, when spot prices were low, the system charged the PCM tank, which increased the overall heating demand at that time.

Three evaluation indicators were used to assess the new strategy: total operational costs, price-responsive heating costs, and peak demand. As shown in Fig. 14, the new strategy effectively reduced peak demand by 11.5 %. The reduction in the price-responsive heating costs was more modest (0.3 %), as the proposed approach prioritized peak shaving based on a setpoint before adjusting charging and discharging actions in response to spot price trends. This explained the smaller shift observed. These two mechanisms, peak demand reduction (which directly reduced peak-related costs) and price-responsive load shifting (which took advantage of spot price variations), together contributed to a 6.1 % reduction in total operational costs compared to the daily time-based strategy.

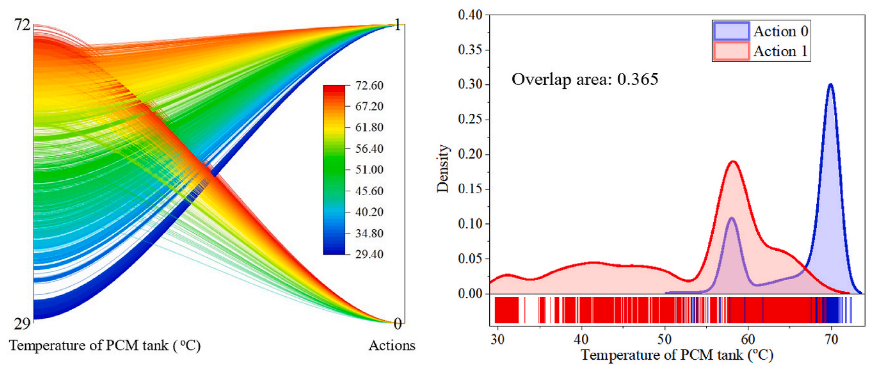
5. Conclusion

This paper presented a methodology for developing a robust SM to enhance the performance of the DH system integrated with a data centre and a PCM tank. Multiple operational strategies for managing the charging and discharging of the PCM tank were implemented in the hybrid system to collect diverse datasets. To investigate the impact of these datasets on the robustness of SMs, LSTM networks were used to train the corresponding SMs, following the identification of appropriate inputs and outputs. Furthermore, a new operational strategy was developed using the best-performing SM to optimize the performance of the hybrid DH system. This strategy focused on peak shaving and load shifting by effectively managing the operations of a PCM tank. The proposed methodology was validated using a case study conducted in Norway.

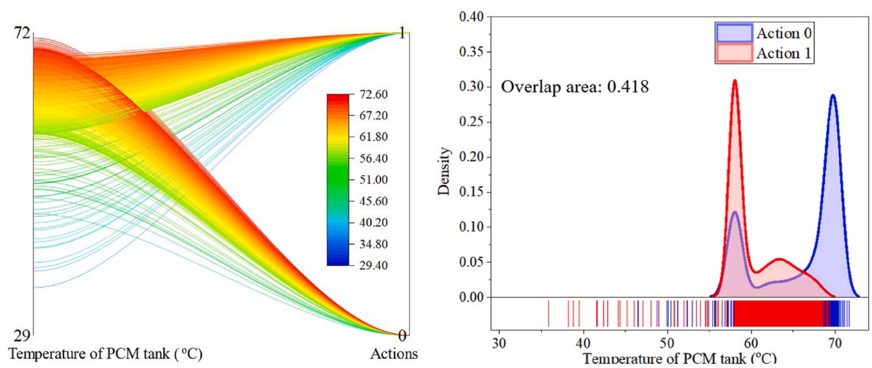
The findings showed that the efficient use of a PCM tank can



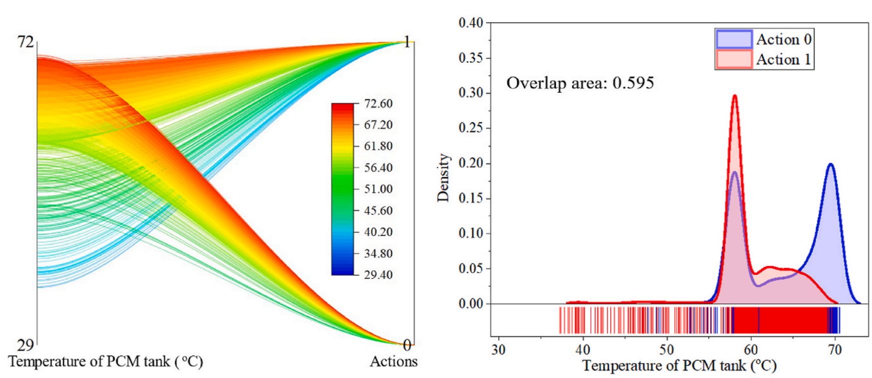
(a) Dataset #1, generated from scenario 1.



(b) Dataset #2, generated from scenario 2.



(c) Dataset #3, generated from scenario 3.



(d) Dataset #4, generated from scenario 4.

Fig. 11. Correlation and density distribution of the PCM tank temperature with actions.

Table 4
Overlap areas of two actions among different datasets.

	Dataset #1	Dataset #2	Dataset #3	Dataset #4
Overlap areas	0.189	0.365	0.418	0.595

significantly reduce operational costs in hybrid DH systems. Datasets generated using different operational strategies significantly influenced the robustness of SMs. Dynamic, day-to-day adjustments in charging and discharging can enhance model robustness, improving system management, and contributing to the creation of well-distributed datasets, ultimately enhancing model performance. Furthermore, a new tailored strategy for managing the PCM tank showed strong effectiveness, reducing peak demand by 11.5 % and price-responsive heating costs by 0.3 %, which contributed to an overall reduction in total operational costs by 6.1 % compared to the baseline.

6. Limitations and future work

This study proposed a methodology for developing a robust SM to enhance the energy flexibility of a DH system integrated with a PCM tank. However, the SMs were developed using TRNSYS simulation data due to the limited availability of the data from real-world systems. Assumptions inherent in the TRNSYS model may restrict the direct applicability of the results. Moreover, the economic feasibility of PCM integration was only partially explored; a comprehensive investment

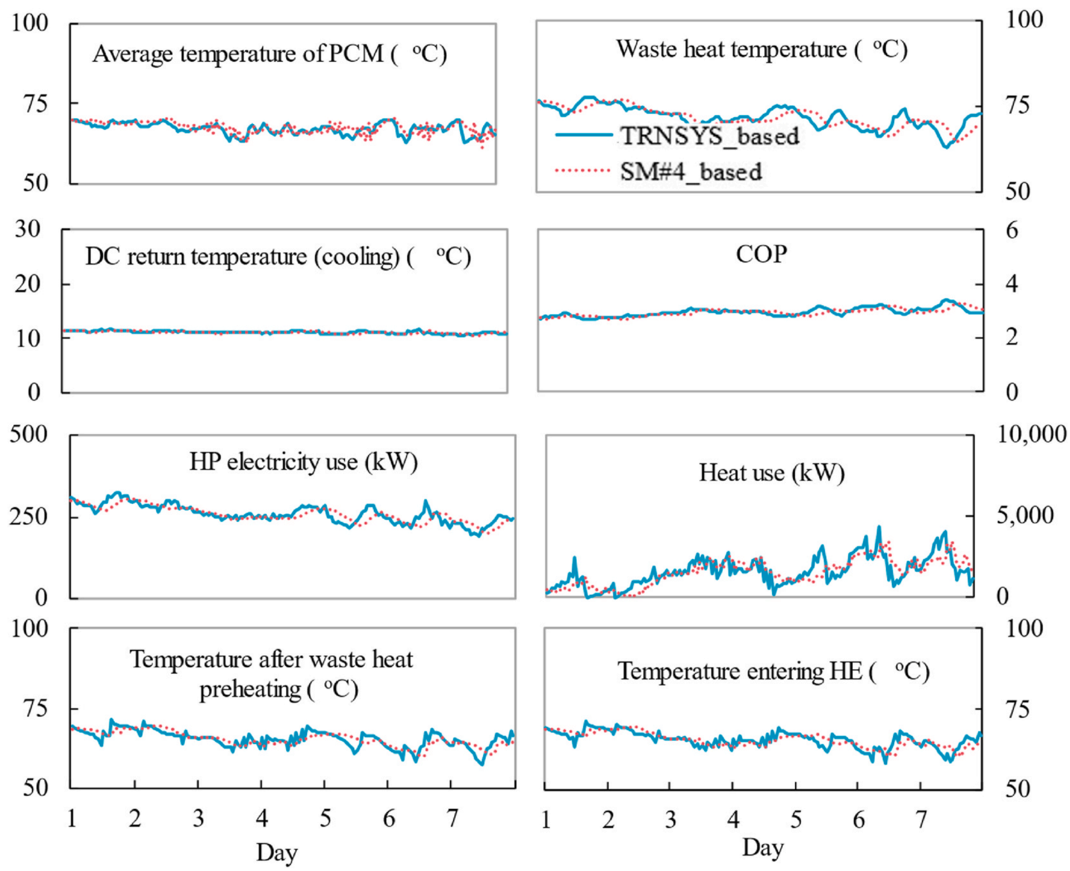


Fig. 12. Comparison of the simulated data from SM #4 with the data generated from TRNSYS simulation over a one-week period.

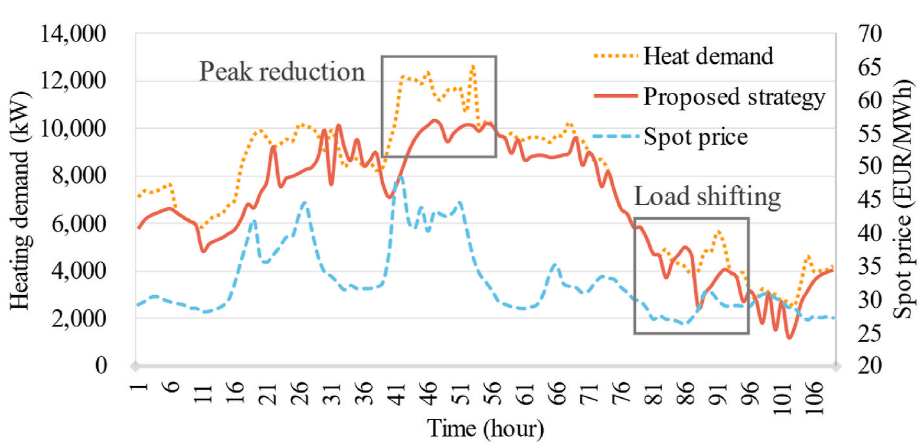


Fig. 13. Flexible heating demand management.

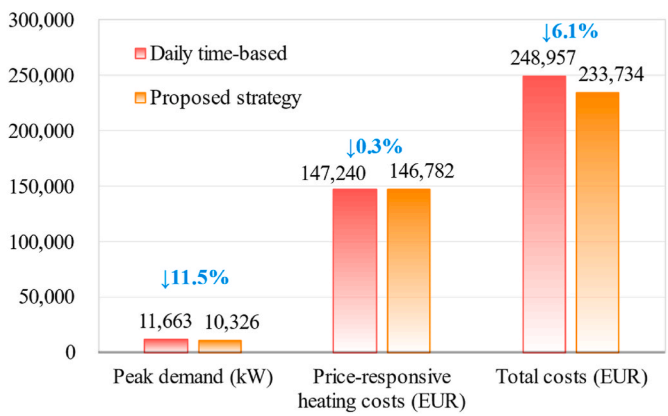


Fig. 14. Comparison of the daily time-based strategy and the proposed strategy.

and cost-benefit analysis is still required. Additionally, the SMs were developed and validated based on a single case study. To apply the methodology to other system configurations or operating conditions, parameter adjustments would be necessary to ensure alignment with the new context.

Future research should validate the methodology across diverse case studies and climatic conditions, assess the robustness of the SMs under data uncertainties, and explore alternative machine learning techniques beyond LSTM to improve model performance and generalizability. Additionally, future work should incorporate experiments with different random seeds and report standard deviations to further strengthen the robustness assessment.

CRediT authorship contribution statement

Han Du: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Xinlei Zhou:** Writing – review & editing, Methodology, Conceptualization. **Natasa Nord:** Writing – review & editing, Supervision, Resources. **Yale Carden:** Writing – review & editing, Supervision, Methodology. **Ping Cui:** Writing – review & editing, Methodology. **Zhenjun Ma:** Writing – review & editing, Supervision, Software, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

Table A1 shows the pseudocode for the training and evaluation process of the LSTM-based surrogate model used in this study.

Table A1
Pseudocode of LSTM-based surrogate model used in this study.

```
Input: Dataset D, sequence length L
Output: Trained LSTM model and evaluation metrics

1. Normalize all features in D to [0, 1]
  2. For each time step t do a. X_t ← D[t: t + L, input features]
  b. Y_t ← D[t + L, output targets]
  3. Split X, Y into training and test sets
  4. Define LSTM model with one LSTM layer and one Dense output layer
  5. Compile model (MSE loss, Adam optimizer, learning rate = 0.01)
  6. Train model with batch size = 128, max epochs = 500, early stopping
  7. For each dataset ∈ {Test} do a. Predict: Ŷ ← model.predict(X)
  b. Reconstruct full-size vectors for inverse scaling
  c. Apply inverse transform to get real values
  d. Compute metrics:
  i. MAE ← (1/n) Σ |(yᵢ − ŷᵢ)/yᵢ|
  ii. RMSE ← sqrt((1/n) Σ (yᵢ − ŷᵢ)²)
  8. Return trained model and metrics
```

Fig. A1 shows the system pipeline diagram of the TRNSYS model used in this study, illustrating the integration of the DH network with a PCM thermal energy storage tank.

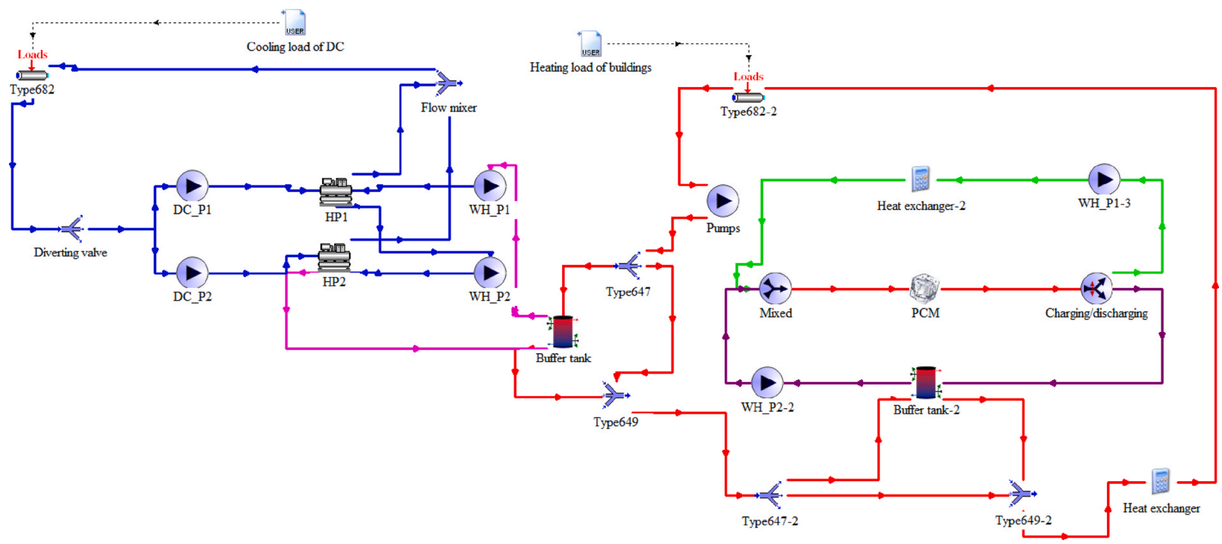


Fig. A1. Pipeline diagram of the TRNSYS model used in this study.

Table A2 summarizes the key terms and performance indicators used in this study, including their definitions and corresponding units.

Table A2
Summary of key terms and indicators used in this study.

Term/Indicator	Definition	Unit
Action 0	Charging action (heat stored in PCM tank)	
Action 1	Discharging action (heat released from PCM tank)	
Spot price	Hourly day-ahead electricity price obtained from the power market	EUR/MWh
Price-responsive heating cost	Heating costs exposed to spot price fluctuations, serving as an economic proxy for load shifting	EUR
Peak demand	Average of the three highest hourly heating demands during the evaluation period	kW
Total operational costs	Overall system costs, including heating and electricity use	EUR

Table A3
RMSE and MAE of surrogate model #4 (reported as mean ± standard deviation over five trainings).

Variables	RMSE	MAE
Average temperature of PCM (^{o}C)	0.558 ± 0.032	0.352 ± 0.03
Waste heat temperature (^{o}C)	0.854 ± 0.026	0.58 ± 0.033
Solid fraction (-)	0.007	0.003
COP (-)	0.112 ± 0.011	0.052 ± 0.002
HP electricity (kW)	7.832 ± 0.045	5.263 ± 0.041
Heat use (kW)	343.305 ± 11.565	223.57 ± 7.282
Temperature after preheating (^{o}C)	1.078 ± 0.029	0.713 ± 0.014

The RMSE and MAE results of surrogate model #4 are reported as mean ± standard deviation across five independent trainings (random seed = 42), as shown in Table A3.

Data availability

Data will be made available on request.

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